Degree and twist estimates for circle-tree invariants

Abstract

We prove a volume asymptotic for the original reduced invariant of an ordered tree of crossing circles. A sufficient condition is that the sum of $d_v/(d_v^2+n_v^2)$, over its circles, is at most 2^{-25}; here d_v is graph degree and n_v is the fixed signed crossing power. Large degree can therefore replace large twist at a vertex. The proof keeps the full finite Newton contraction, including its repeated-root derivatives, and estimates all discarded indices before forming a complex wire sum. A joint Fourier argument then controls all charges, endpoint currents and connecting contours. Its first charge has a nonzero coefficient through both level parities. The geometric circle-tree theorem identifies the resulting action with the volume of the original filled knot complement. We also give the centered microscopic main-meridian profile, with the physical framing and edge factors retained.

1 The diagrams and the conclusion

A crossing circle is an auxiliary component bounding a disk punctured by two main strands. Filling it restores an integer power of the crossing in that disk's original frame. Consider the following literal ordered construction, as in [2, Section 1]. A finite tree $\mathcal{T}=(V,E)$ has at least one edge. Every vertex has its own crossing-circle tangle P_v. A rooted expression is

$$W_v=\left(\prod_{w\in I_v^-}\mathsf{T}(W_w)\right)P_v\left(\prod_{w\in I_v^+}\mathsf{T}(W_w)\right). \tag{1.1}$$

The lists have their specified orders. The turn rotates the whole tangle and its disk frame; it is not just a scalar change of variable. Choose a terminal root and the prescribed cylindrical closure. Keep the actual strand orientations, a reduced cut on a main strand, and zero framing. Require that the endpoint tracing gives one main component and no internal closed component. There are no bare product junctions. This is an input condition on the diagram, not a consequence of the abstract graph.

Write $r=|V|$, $e=|E|=r-1$, and $d_v\geq 1$ for graph degree. At vertex v the physical crossing power is $n_v\in\mathbb{Z}$. The diagram, all powers and orientations are fixed throughout. Only the level $N\geq 3$ tends to infinity. Let $F_N(M)$ be its reduced zero-framed ADO invariant, with unknot value one, in the physical convention of [1]. At $M=1$ it is the reduced Kashaev invariant of the same knot $K_{\mathbf{n}}$. Put

$$q_N=e^{\pi\mathrm{i}/N},\qquad d=N-1,\qquad M=q_N^s,\qquad |s|\leq R<1.$$

The level degree d is unrelated to d_v.

Here is the quantitative hypothesis:

$$\mathfrak{S}=\sum_{v\in V}\frac{d_v}{d_v^2+n_v^2}\leq 2^{-25}. \tag{1.2}$$

Neither the number of vertices nor the signs or relative sizes of the fixed powers are bounded. A power can be zero when its degree is large enough. The constants in the asymptotic and the initial level may depend on the whole fixed diagram.

Define $J(x)=\int_0^x \log(2\sin \pi u)\,\mathrm{d}u$, $g(x)=J(2x)-4J(x)$, and $G_0=g(1/4)$. Near $t=u=0$ let

$$\psi(0,0)=2G_0, \qquad \psi_t(t,u)=-\frac{1}{\pi}\operatorname{arsinh}(\tan(t/2)\cos(u/2))\,, \qquad \psi_u(t,u)=\psi_t(u,t). \tag{1.3}$$

Branches continue from zero. The mixed derivatives agree, as checked in Section 3. Set

$$\Psi(\boldsymbol{\tau})=\sum_{vw\in E}\psi(\tau_v,\tau_w), \qquad F_0=2eG_0, \qquad \Phi_{\boldsymbol{\ell}}(\boldsymbol{\tau})=\Psi(\boldsymbol{\tau})+\frac{\mathrm{i}}{4\pi}\sum_v n_v\tau_v^2-\mathrm{i}\sum_v \ell_v\tau_v.$$

Let $\boldsymbol{\tau}_*$ be the small critical point for $\boldsymbol{\ell}=\mathbf{1}$, and $\mathcal{C}(\mathbf{1})=\Phi_{\mathbf{1}}(\boldsymbol{\tau}_*)$. Existence and uniqueness in the needed chart are proved below.

Theorem 1.1. *For the fixed marked knot diagram above, suppose (1.2). There is a constant $D_{\mathcal{T},\mathbf{n}}\neq 0$, independent of the level parity, such that*

$$F_N(1)=\Omega_N N^{3/2} D_{\mathcal{T},\mathbf{n}} e^{N\mathcal{C}(\mathbf{1})}(1+O(N^{-1})), \qquad \Omega_N=q_N^{d^2(2w(D)+\sum_v n_v)/4}. \tag{1.4}$$

Here $w(D)$ is the oriented writhe of the original diagram and $|\Omega_N|=1$. The original knot complement is hyperbolic and

$$\lim_{N\to\infty}\frac{2\pi}{N}\log|F_N(1)|=2\pi\,\mathrm{Re}\,\mathcal{C}(\mathbf{1})=\mathrm{Vol}(S^3\setminus K_{\mathbf{n}}). \tag{1.5}$$

Both level parities are included.

For the centered profile put

$$H_N(s)=\frac{\sin \pi s}{N\sin(\pi s/N)}, \qquad A_N(s)=q_N^{(-w(D)-\sum_v n_v)s^2/2}.$$

The value at zero is removable. For each edge let $x^*_{vw}=x(\tau_{*,v},\tau_{*,w})$, where the branch $x(0,0)=1/4$ is specified by

$$\tan^2(\pi x(t,u))=\frac{1}{1+\tan^2(t/2)+\tan^2(u/2)}. \tag{1.6}$$

Theorem 1.2. *Under the same hypothesis, $F_N(q_N^s)$ is nonzero on $|s|\le R<1$ for all sufficiently large N, and uniformly there*

$$\frac{F_N(q_N^s)}{A_N(s)H_N(s)F_N(1)}=1-\frac{\pi s^2}{N}\sum_{vw\in E}\cot(\pi x^*_{vw})+O_R(N^{-2}). \tag{1.7}$$

Consequently the limit in (1.5) also holds uniformly with $F_N(1)$ replaced by $F_N(q_N^s)$.

This is a microscopic statement: s is bounded while $M=q_N^s$ approaches one. It asserts neither an asymptotic on a fixed M-neighborhood nor an all-orders annihilator theorem.

The finite inputs are precisely [1, Proposition 6.1, Theorem 7.1 and Proposition 8.1]: full crossing polynomials and traces, the original tree contraction, and its scalar ledger. The geometric input is [2, Theorem 1.1]. The analytic passage between them is proved here. In particular the independently long-twist theorem of [4] is not applicable merely from (1.2); a vertex may have $n_v=0$.

2 The original finite contraction and its discarded indices

2.1 Fixed coordinates and the physical factors

Write $\theta = \pi/N$, $\eta = (-1)^{N-1}$, and let $\mathsf{T}_N(z)$ be the monic polynomial with $\mathsf{T}_N(t+t^{-1}) = t^N + t^{-N}$. The pair algebra at a vertex has relation $P_\sigma = \mathsf{T}_N + 2\cos(2\pi\sigma)$, with $\sigma = 0, s$, or $-s$ prescribed by its oriented pair type. The common vector-space basis is

$$e_k(z) = S_k^{-2} \prod_{j<k} (c_j - z), \qquad S_k = \prod_{j=1}^{k} 2\sin(j\theta), \qquad c_j = 2\cos((2j+1)\theta), \quad 0 \le k < N. \tag{2.1}$$

An empty product is one. A canonical representative has degree less than N, obtained using its actual relation P_σ. The different relations are not identified as algebras.

The closing functional and paired edge factors are

$$\phi_\sigma(f) = \eta N^2 [z^d] \operatorname{rem}_{P_\sigma} f, \qquad \phi_\sigma(e_i e_j) = \mathbf{1}_{i+j=d}, \tag{2.2}$$

$$W_k(s) = \prod_{j=1}^{k} \left(1 - \frac{\sin^2(\pi s/N)}{\sin^2(\pi j/N)}\right), \qquad W_k W_{d-k} = H_N^2. \tag{2.3}$$

Every physical turn contributes its prescribed scalar $\gamma_e(s) \in \{1, q_N^{s^2/2}, q_N^{-s^2/2}\}$. Lowering its index with (2.2) identifies the equal indices on the two ends of the edge and leaves $\gamma_e H_N/W_k$, not its reciprocal.

Let p_v be the actual crossing power, including the carrier specified by the original oriented diagram. The exact identity of [1, Theorem 7.1] is

$$\begin{aligned} Z_N(s) &= \sum_{\boldsymbol{k} \in \{0,\dots,d\}^E} \prod_v \phi_{\sigma_v} \left(p_v \prod_{e \ni v} e_{k_e} \right) \prod_e \gamma_e(s) \frac{H_N(s)}{W_{k_e}(s)}, \\ F_N(q_N^s) &= \vartheta_{d+s}^{-w(D)} H_N(s) Z_N(s), \qquad \vartheta_{d+s} = q_N^{(s^2-d^2)/2}. \end{aligned} \tag{2.4}$$

Thus cups, caps, physical turns, negative crossings, the pivotal closure and the one reduced cut are already fixed finite data.

On every disk $R < 1$, all factors H_N/W_k and their inverses are bounded independently of N, k. For example split the factors in (2.3) according to $\min(j, N-j) \le J$ or $> J$. The latter logarithms are bounded by a summable constant times $\min(j, N-j)^{-2}$. For fixed J the former factors have no zeros on the disk; their endpoint limits are $1 - s^2/j^2$, also zero-free. Compactness, including the finitely many small N, gives the bound and its inverse. The same argument applies to H_N.

2.2 Bounds that retain the first normal

Let $m = \lfloor d/2 \rfloor$. Indices $k \le m$ will be called low, the others high. For a canonical polynomial use the complete norm

$$\|f\|_{\mathcal{J}} = \max\{|f(c)|,\ |f'(c)| :\ c \text{ a double node of } P_0;\ |f(-2)| \text{ if } N \text{ is odd}\}.$$

The derivative at the odd simple node is absent. All first derivatives at double nodes are retained.

Lemma 2.1. *For fixed $R < 1$, a fixed number of factors and fixed integer crossing powers, quotient multiplication, closing, and those crossing powers have polynomial cost in $\|\cdot\|_{\mathcal{J}}$. Uniformly in all indices,*

$$\|e_k\|_{\mathcal{J}} \le C N^A \begin{cases} \exp(N g(k/N)), & 0 \le k \le m, \\ 1, & m < k < N. \end{cases} \tag{2.5}$$

The constants in these statements do not depend on s in its disk.

Proof. We give the conditioning and endpoint details because ignoring derivatives would make the high-index assertion insufficient. The distinct complete-node polynomial is

$$Q_N(z) = \begin{cases} 2T_{N/2}(z/2), & N \text{ even}, \\ (z+2)\{U_m(z/2) - U_{m-1}(z/2)\}, & N = 2m+1. \end{cases}$$

Here T_j, U_j are the ordinary Chebyshev polynomials. Its interval norm and derivatives are polynomial in N. At its roots, $|Q_N'| \geq N/2$ in the even case and at least $N/2$ in the odd case: for an odd interior root $z = 2\cos t$, $|Q_N'(z)| = N/(2\sin(t/2))$, while $|Q_N'(-2)| = N$. The mean-value formula bounds each Lagrange cardinal by a polynomial in N.

For a general polynomial of degree $< N$, interpolate its values by I of degree $< \deg Q_N$. Write $f - I = Q_N h$. The values of h at the double nodes are $(f'(c) - I'(c))/Q_N'(c)$. These determine h, whose degree is less than their number. Use Q_N for this interpolation at even levels and $Q_N/(z+2)$ at odd levels; the same cardinal bound applies. Consequently the interval norm, the complete normal norm, and the Chebyshev coefficient sum are polynomially equivalent. One may bound derivatives without any conditioning assumption: a degree-n Chebyshev expansion has coefficients at most twice its interval norm, and $\|T_j'(z/2)\| \leq j^2/2$.

Write $C_j(z) = (t^j + t^{-j})/2$ when $z = t + t^{-1}$. The product identity $2C_iC_j = C_{i+j} + C_{|i-j|}$ makes the coefficient sum submultiplicative. Modulo P_σ,

$$C_N = -\cos(2\pi\sigma), \qquad \operatorname{rem} C_{N+j} = -2\cos(2\pi\sigma)C_j - C_{N-j} \quad (0 < j < N).$$

Iteration bounds reduction of degree at most hN by $(2\cosh(2\pi R) + 1)^h$, for fixed h. Finally $[z^d]C_d = 1/2$. These facts prove the asserted polynomial costs of multiplication and ϕ_σ.

The actual crossing polynomial and its inverse are holomorphic through the collisions in $|\sigma| < 1$, with the normals of [1, Proposition 6.1]. On a parameter circle of radius $R' \in (R, 1)$, $R' \neq 1/2$, Lagrange interpolation has denominator $\sin(2\pi\sigma)$ bounded away from zero. Its nodes have angular imaginary part $O_{R'}(N^{-1})$. The divided differences of T_N and their derivatives on that ellipse are $O_{R'}(N^2)$ and $O_{R'}(N^4)$: differentiate $2\cos(Nt)$, with endpoint limits interpreted by its polynomial expression. The fixed-power eigenvalues, including negative powers, are bounded there. Thus the canonical crossing and its first derivative have polynomial interval norm. Holomorphy and the maximum principle extend the bound through all collisions to the disk. The mixed orders use their actual fixed reference crossing and the bounded scalars of the same proposition, so they satisfy the same conclusion.

For low k, all values $e_k(c_j)$ at distinct nodes are nonnegative and bounded by $e_k(-2)$. Interpolation by Q_N therefore gives $\|e_k\|_{\mathcal{J}} \leq N^A e_k(-2)$. Uniform sum–integral comparison of the midpoint cosine product and sine factorial gives

$$\log e_k(-2) = Ng(k/N) + O(\log N), \qquad 0 \leq k \leq m.$$

This remains uniform at both endpoints: removing the first and last mesh interval leaves a smooth logarithm, while each removed logarithmic interval contributes $O(\log N)$.

For high k, put $l = N - k \leq \lfloor N/2 \rfloor$ and $B_j(z) = \prod_{a<j}(c_a - z)$. Reflection $c_{d-a} = c_a$ gives $B_kB_l = (-1)^N P_0$. Thus e_k vanishes at every node. At a double node indexed by $0 \leq j < \lfloor N/2 \rfloor$, its derivative is zero unless $j < l$, and otherwise

$$|e_k'(c_j)| = \frac{S_{l-1}^2}{2\sin((2j+1)\theta)S_{l+j}S_{l-1-j}}. \tag{2.6}$$

To obtain this formula, compare quadratic coefficients in $B_kB_l = (-1)^N P_0$, using $P_0''(c_j)/2 = N^2/(4\sin^2((2j+1)\theta))$, $|B_l'(c_j)| = S_{l+j}S_{l-1-j}/(2\sin((2j+1)\theta))$, and $S_kS_{l-1} = N$. The

denominator ratio $D_j = S_{l+j}S_{l-1-j}/S_{l-1}^2$ starts at $2\sin(l\theta)$ and is nondecreasing, since

$$\frac{D_{j+1}}{D_j} = \frac{\sin((l+j+1)\theta)}{\sin((l-1-j)\theta)} \geq 1.$$

The two angles sum to at most π. Equation (2.6) is consequently at most $1/(4\sin((2j+1)\theta)\sin(l\theta)) \leq N^2/16$. The odd simple-node value is zero. This proves (2.5). □

2.3 Discard before forming an angular sum

Retain $k_e/N \in I = [1/5, 3/10]$ at every edge. Each edge index occurs in two vertex factors of (2.4). Lemma 2.1 therefore bounds the absolute rate of an assignment by $2\sum_{\text{low } e} g(k_e/N)$. Direct differentiation gives

$$g'(1/4) = 0, \qquad g''(x) = -\frac{4\pi}{\sin 2\pi x} \leq -4\pi.$$

Since $g(0) = g(1/2) = 0$, concavity also gives $2G_0 \geq \pi/4$. A low index outside I loses at least $\pi/100$ from F_0; a high index loses at least $2G_0$. The number of assignments is polynomial in N for the fixed tree. We have proved the full-source estimate

$$|\text{all discarded assignments}| \leq C_R N^A e^{N(F_0-\eta_0)}, \qquad \eta_0 = \pi/200. \tag{2.7}$$

The exponent loss is independent of tree size and powers; the prefactor and starting level are not. This argument estimates the actual vertex traces, rather than replacing a small spectral norm by a positive coefficient cone.

On the retained indices, the edge sums commute with the polynomial traces. Remove only the already recorded edge scalars and put

$$\begin{aligned} K_N(t,u;s) &= \sum_{k/N\in I} e_k(-2\cos t)e_k(-2\cos u)\frac{H_N(s)}{W_k(s)}, \\ B_N(\boldsymbol{\tau},s) &= \prod_{vw\in E} K_N(\tau_v,\tau_w;s). \end{aligned} \tag{2.8}$$

This is an exact polynomial in each $-2\cos\tau_v$. Its separate degree may be larger than $N-1$; no such bound is assumed in the contours below. The functional ϕ_{σ_v} reduces the whole polynomial in its proper relation.

3 An exact complex edge sum

3.1 The interpolation and its normalization

We use the canonical factorial lemma [3, Lemma 2.1], whose statement we record. On $-1 < \operatorname{Re} z < N$,

$$G_N(z) = \left(\frac{2\pi}{N}\right)^z \Gamma(z+1)\prod_{a\geq 1}\frac{\Gamma(Na+z+1)}{\Gamma(Na-z)(Na)^{2z+1}} \tag{3.1}$$

is holomorphic and nonzero, with positive-base powers. It satisfies $G_N(0) = 1$, $G_N(k) = S_k$, and

$$\frac{G_N(z+1)}{G_N(z)} = 2\sin\frac{\pi(z+1)}{N}, \qquad G_N(z)G_N(d-z) = N, \qquad G_N(s)G_N(-s) = H_N(s)^{-1}.$$

On every compact subset of $0 < \operatorname{Re} x < 1$,

$$\log G_N(Nx) = NJ(x) + \tfrac{1}{2}\operatorname{Log}(2N\sin\pi x) + \frac{\pi\cot\pi x}{12N} + O(N^{-3}). \tag{3.2}$$

All fixed derivatives, and any fixed further inverse-power order, are uniform on smaller compact sets. Logarithms and square roots are continued from the positive real interval. This numbered lemma includes the convergent product and summable Stirling proof; no periodic factor may be inserted into (3.1). For background on those classical expansions see [5, Section 5.11].

The shift equation now proves the exact interpolation needed here. For $u = Nt/(2\pi)$, define

$$\mathcal{B}_N(x,t) = \frac{G_N(d/2+u)G_N(d/2-u)}{G_N(d/2-Nx+u)G_N(d/2-Nx-u)G_N(Nx)^2}. \tag{3.3}$$

At $x = k/N$ this equals $e_k(-2\cos t)$. Indeed successive quotients are the original cosine numerator and sine-factorial denominator of (2.1). The exact edge insertion is

$$\mathcal{I}_N(x,s) = \frac{G_N(Nx)^2}{G_N(Nx-s)G_N(Nx+s)}, \qquad \mathcal{I}_N(k/N,s) = \frac{H_N(s)}{W_k(s)}. \tag{3.4}$$

The second identity follows by iterating the same shift equation and using $G_N(s)G_N(-s) = H_N^{-1}$.

Put

$$b(x,t) = \int_0^x \operatorname{Log}(2\cos(2\pi v) + 2\cos t)\,\mathrm{d}v - 2J(x). \tag{3.5}$$

Branches start at real $x \in I, t = 0$. On a fixed complex tube about I, (3.2) gives

$$\begin{aligned} \mathcal{B}_N(x,t) &= \frac{e^{Nb(x,t)}}{2N\sin\pi x}(1 + N^{-1}b_1(x,t) + O(N^{-2})), \\ \mathcal{I}_N(x,s) &= 1 - \frac{\pi s^2}{N}\cot\pi x + O_R(N^{-2}). \end{aligned} \tag{3.6}$$

The first correction b_1 is independent of s. To check the leading constant, each argument $d/2 \pm u$ has the shift $-1/2$, so its square-root factor is $\sqrt{N}$; the analogous denominator factors cancel these, leaving $(2N\sin\pi x)^{-1}$. Thus no midpoint shift has been discarded. For the second identity use $(\log G_N)''(Nx) = \pi\cot(\pi x)/N + O(N^{-2})$ in the symmetric difference. All remainders are holomorphic with uniform fixed derivatives on a smaller tube.

3.2 The exact stationary phase and an available contour

For $A(x,t,u) = b(x,t) + b(x,u)$, its critical branch satisfies

$$\cos(2\pi x) = \frac{1 - \cos t\cos u}{2 + \cos t + \cos u}, \qquad x(0,0) = 1/4. \tag{3.7}$$

This is equivalent to (1.6). It follows by writing $A_x = 0$, squaring the products of the nonzero factors, and choosing the branch at zero; the original unsquared equation holds there and hence nearby. Substitute $z = \tan\pi v$ into the t-derivative of (3.5). At its critical x, the resulting logarithmic antiderivative is $-\pi^{-1}\operatorname{arsinh}(\tan(t/2)\cos(u/2))$. Thus $\psi = \operatorname{stat}_x A$ is exactly (1.3). In particular ψ is separately even.

We need a domain for the finite sum, not just this limiting phase. On $|t|, |u| \le 1/4$, $|\tan(t/2)| < 0.128$; the tangent form of (3.7) implies

$$|\cos(2\pi x)| < 0.017, \qquad |x(t,u) - 1/4| < 0.003.$$

On $\operatorname{Re} x \in I$, $|\operatorname{Im} x| \le 1/50$, direct differentiation gives

$$A_{xx} = \sum_{\xi=t,u}\left[-\frac{2\pi\sin 2\pi x}{\cos 2\pi x + \cos\xi} - 2\pi\cot\pi x\right], \qquad |A_{xx}| < 44, \quad \operatorname{Re} A_{xx} < -12. \tag{3.8}$$

Here are sufficient elementary bounds: $|\cos 2\pi x| < 0.338$, $|\cos\xi - 1| < 0.032$, $|\sin 2\pi x| < 1.009$, and $\operatorname{Re}\cot \pi x > 0.72$. The real part of $\sin 2\pi x/(\cos 2\pi x + \cos\xi)$ exceeds 0.328: its conjugated numerator is at least $0.95(0.655) - 0.040(0.159)$, and its squared denominator is at most 1.37^2. These also prove the modulus bound.

Enclose precisely the retained integers k by a rectangle in the x-plane with vertical half-grid sides and horizontal height $h = 1/500$. The residue formula with $\pi N \cot(\pi N x)$ equals the finite sum exactly. On a vertical half-grid side the cotangent has modulus at most one. On horizontal sides it equals $\mp\mathrm{i} + O(e^{-2\pi N h})$. If $X = \operatorname{Re}(x - x_*), Y = \operatorname{Im}(x - x_*)$, Taylor's formula and (3.8) imply

$$\operatorname{Re}\{A(x) - A(x_*)\} \le -6X^2 + 44|XY| + 22Y^2. \tag{3.9}$$

On a vertical side, for all large N, $|X| \ge 0.047 - o(1)$ and $|Y| \le 0.005$; the right side is below $-0.002 + o(1)$. On a horizontal side its maximum in X is at most $(308/3)Y^2 \le 77/30000$, less than $2\pi h = \pi/250$. All nonconstant cotangent modes and vertical currents therefore have a fixed exponential loss relative to $e^{N\psi}$.

The constant horizontal terms give N times a real integral. Translate this integral to the real tangent through x_*. Its endpoint connectors have the same loss, and $\operatorname{Re} A_{xx} < -12$ gives a nondegenerate Gaussian. The ordinary Taylor proof of the saddle expansion applies uniformly: outside a fixed small saddle interval there is a fixed loss; inside it put $x - x_* = N^{-1/2}z$, expand, and integrate the Gaussian-dominated remainder. Odd terms integrate to zero. This also gives any fixed order and its parameter derivatives. All Gamma arguments in (3.3) stay in regular bulk strips throughout these contours.

It follows that

$$\begin{aligned} K_N(t,u;0) &= N^{-3/2} a_e(t,u) e^{N\psi(t,u)}(1 + N^{-1}a_{e,1}(t,u) + O(N^{-2})), \\ a_e &= \frac{\sqrt{2\pi}}{4\sin^2(\pi x_*)\sqrt{-A_{xx}(x_*,t,u)}}. \end{aligned} \tag{3.10}$$

The square root is positive at zero and continued on the disk; no factor vanishes there. In particular $a_e(0,0) = 1/4$. Applying the same contour to (3.4) proves

$$\begin{aligned} B_N(\boldsymbol{\tau},0) &= N^{-3e/2} a(\boldsymbol{\tau}) e^{N\Psi(\boldsymbol{\tau})}(1 + N^{-1}a_1(\boldsymbol{\tau}) + O(N^{-2})), \\ \frac{B_N(\boldsymbol{\tau},s)}{B_N(\boldsymbol{\tau},0)} &= 1 - \frac{\pi s^2}{N}\sum_{vw\in E}\cot(\pi x(\tau_v,\tau_w)) + O_R(N^{-2}), \qquad a = \prod_e a_e. \end{aligned} \tag{3.11}$$

For a holomorphic insertion, its first saddle correction is $O(N^{-1})$, since the linear Gaussian term is odd. Multiplication by its prefactor $1/N$ in (3.6) therefore makes the last remainder $O(N^{-2})$. The common a_1 cancels in this ratio.

4 Angular bounds and weighted curvature

4.1 A bound on the entire real angular period

Let $N \ge 160$, $k/N \in I$, and $r_j = \cos((2j+1)\pi/N)$. The exact product is

$$\frac{e_k(-2\cos z)}{e_k(-2)} = \prod_{j<k} Q_{r_j}(z), \qquad Q_r(z) = \frac{r + \cos z}{1 + r}.$$

Reserve the first $m_0 = \lfloor N/32 \rfloor$ factors, where $r_j > \cos(\pi/16) > 9/10$. There are at most $N/20 + 1$ negative parameters, all $> -2/5$. Pair them injectively with parameters among the

next $N/20+1$ positions, which are $>7/10$; these positions are disjoint from the negative ones since $m_0+N/20+1\le N/8$. All remaining unpaired parameters are nonnegative.

For a pair $r>-2/5, s>7/10$, the coefficients of

$$(z^2+2rz+1)(z^2+2sz+1)$$

are $1, 2(r+s), 2+4rs, 2(r+s), 1$, all positive. Normalized at $z=1$, with its central Laurent power removed, it is a symmetric probability measure on $\{-2,-1,0,1,2\}$. A remaining single factor has the analogous interpretation on $\{-1,0,1\}$. Its complex-angular modulus is at most $\cosh(by)\le e^{b^2y^2/2}$, where b is its Laurent degree.

For a reserved factor, put $w=\sin^2(t/2)$. The two linear pieces of $|1-2w/(1+r)|$ give $|Q_r(t)|\le 1-w/2$. For $|y|\le 1/4$,

$$|r+\cos(t+\mathrm{i}y)|^2=(r+\cos t)^2+2r\cos t(\cosh y-1)+\sinh^2 y.$$

Since $2(\cosh y-1)+\sinh^2 y\le 4y^2$, each reserved factor is bounded by $\exp(-w/4+2y^2)$. Use $m_0\ge N/40$, then multiply the paired and reserved estimates:

$$\left|\frac{e_k(-2\cos(t+\mathrm{i}y))}{e_k(-2)}\right|\le \exp N\left(-\frac{\sin^2(t/2)}{160}+4y^2\right). \tag{4.1}$$

This pairing precedes absolute values. Individual Laurent factors need not have nonnegative coefficients.

Set $a_v=d_v/(2\pi)$. On $|t_v|\le\pi,\ |y_v|\le 1/4$, $\sin^2(t_v/2)\ge t_v^2/\pi^2$, so (4.1), the bounded edge insertions, and the positive zero-angle sum give

$$|B_N(\mathbf{t}+\mathrm{i}\mathbf{y},s)|\le C_R N^A\exp N\left\{F_0-\frac{1}{400}\sum_v a_v t_v^2+70\sum_v a_v y_v^2\right\}. \tag{4.2}$$

The stronger incidence constants are $1/(80\pi)>1/400$ and $8\pi<70$. Endpoint translations $O_R(N^{-1})$ and derivatives cost only polynomial factors for the fixed tree; one may apply Cauchy's formula on radius $1/N$ disks. This is a bound on the entire angular period, not only near its maximum.

4.2 The local nonlinear terms

With $Q=\sqrt{1-\sin^2(t/2)\sin^2(u/2)}$, differentiation gives

$$\psi_{tt}=-\frac{\cos(u/2)}{2\pi\cos(t/2)Q},\qquad \psi_{tu}=\frac{\sin(t/2)\sin(u/2)}{2\pi Q},\qquad \det\psi''=\frac{1}{4\pi^2}. \tag{4.3}$$

Consequently

$$\Psi=F_0-\frac{1}{2}\sum_v a_v\tau_v^2+\mathcal{E}(\boldsymbol{\tau}).$$

On the polydisk $|\tau_v|\le 1/4$, put $\delta=1/50,\ \varepsilon=1/25$. Then

$$\left|\frac{\mathcal{E}_v}{\tau_v}\right|\le\delta a_v,\qquad \sum_w|\mathcal{E}_{vw}|\le\varepsilon a_v,\qquad |\mathcal{E}|\le\frac{\varepsilon}{2}\sum_v a_v|\tau_v|^2. \tag{4.4}$$

The quotient extends at zero by separate evenness. For explicit bounds write $k=\cosh(1/8)-1<0.008$, $\sigma=\sinh(1/8)<0.126$, and $C=(1-\sigma^4)^{-1/2}$. The normalized diagonal error in (4.3) is at most $2kC/(1-k)+C-1<1/50$, and the off-diagonal bound is σ^2C; their sum is $<1/25$. Summing over incident edges proves the row estimate. Integrate the diagonal error while fixing the other variables to prove the first estimate. For the last, sum the edge quadratic forms and use $2|zw|\le|z|^2+|w|^2$. No estimate grows with the number of vertices.

5 Critical points with degree in the denominator

Set

$$b_v = \frac{n_v}{2\pi}, \qquad D_v = a_v - \mathrm{i}b_v, \qquad t_0 = \frac{1}{16}, \qquad L_v = \lfloor t_0|D_v| \rfloor. \tag{5.1}$$

The summed condition implies $\sqrt{d_v^2 + n_v^2} \geq 2^{25/2}\sqrt{d_v}$, so $|D_v| > 32$ and $L_v \geq 1$. This denominator stays large also when $n_v = 0$.

For real continuous charges $0 \leq \ell_v \leq L_v$, the critical equation of $\Phi_{\boldsymbol{\ell}}$ is

$$D_v\tau_v = -\mathrm{i}\ell_v + \mathcal{E}_v(\boldsymbol{\tau}). \tag{5.2}$$

After division by D_v, its right side is a contraction with constant at most ε in the radius-$1/4$ polydisk, by (4.4) and $a_v \leq |D_v|$. Its image has radius at most $t_0 + \delta/4 < 1/4$. It has a unique analytic critical branch $\boldsymbol{\tau}_{\boldsymbol{\ell}}$. Define $\mathcal{C}(\boldsymbol{\ell}) = \Phi_{\boldsymbol{\ell}}(\boldsymbol{\tau}_{\boldsymbol{\ell}})$.

Put $Q_v = \mathcal{E}_v/\tau_v$, with its holomorphic removable extension. The exact equation is

$$\tau_v = -\frac{\mathrm{i}\ell_v}{D_v - Q_v}. \tag{5.3}$$

Since $|Q_v| \leq \delta a_v$, define

$$\gamma_+ = \frac{1+\delta}{(1-\delta)^2} = \frac{2550}{2401}, \qquad \gamma_- = \frac{1-\delta}{(1+\delta)^2} = \frac{2450}{2601}.$$

Taking the imaginary part of (5.3) gives

$$\gamma_- \frac{\ell_v a_v}{|D_v|^2} \leq -\operatorname{Im}\tau_v \leq \gamma_+ \frac{\ell_v a_v}{|D_v|^2}. \tag{5.4}$$

For example the numerator is $\ell_v \operatorname{Re}(D_v - Q_v)$, between $(1-\delta)\ell_v a_v$ and $(1+\delta)\ell_v a_v$; the denominator is bounded by $(1 \pm \delta)^2|D_v|^2$. Every positive-charge coordinate therefore has negative imaginary part and is nonzero.

The envelope derivative is $\partial_{\ell_v} \operatorname{Re}\mathcal{C} = \operatorname{Im}\tau_v$. Integrating it one coordinate at a time gives

$$\operatorname{Re}\mathcal{C}(\boldsymbol{\ell}) \leq \operatorname{Re}\mathcal{C}(\mathbf{1}) - \frac{\gamma_-}{2}\sum_v \frac{a_v}{|D_v|^2}(\ell_v^2 - 1), \qquad 1 \leq \ell_v \leq L_v. \tag{5.5}$$

Integrating from $\mathbf{0}$ to $\mathbf{1}$ similarly gives

$$F_0 - \beta \leq \operatorname{Re}\mathcal{C}(\mathbf{1}) \leq F_0, \qquad \beta = \frac{\gamma_+}{2}\sum_v \frac{a_v}{|D_v|^2} = \pi\gamma_+\mathfrak{S}. \tag{5.6}$$

This compares each charge directly with the filled first-charge action; a comparison only with F_0 would not suffice.

The real part of the Hessian on a real tangent is at most $-(1-\varepsilon)\operatorname{diag}(a_v)$, by the symmetric weighted row bound in (4.4). Its Gaussian determinant is nonzero. Also $|\tau_v| \leq t_0/(1-\delta) < \pi/2$, so $\prod_v \sin\tau_v \neq 0$ at every positive-charge saddle.

6 The entire Fourier filling calculation

6.1 A translated finite grid, including its endpoint currents

For an even trigonometric polynomial $f(t) = f^\sharp(-2\cos t)$, define the exact trace $\mathcal{D}_{n,\sigma,N}[f] = \phi_\sigma(R_\sigma^n f^\sharp)$. There is no restriction on the degree of $f^\sharp$. The actual crossing formula of [1, Proposition 6.1 and equation (6.4)] gives

$$\mathcal{D}_{n,\sigma,N}[f] = \frac{\eta N C_{N,n}(\sigma)}{\sin(2\pi\sigma)} \sum_{j=0}^{N-1} \sin\tau_j \, f(\tau_j) e^{\mathrm{i}nN\tau_j^2/(4\pi)}, \qquad \begin{cases} \tau_j = \dfrac{2\pi}{N}(d/2 + \sigma - j), \\ C_{N,n}(\sigma) = q_N^{n(d^2/4 - \sigma^2/2)}. \end{cases} \tag{6.1}$$

It is initially written off the colliding grids. Its holomorphic limit is the full trace, including first normals and the odd simple-node value; it is not a distinct-node approximation.

Fix $R < R' < 1$, with $R' \neq 1/2$, and first put the formal parameters σ_v on $|\sigma_v| = R'$. Their denominators are then bounded away from zero. The grid kernel

$$\mathcal{K}_\sigma(t) = \frac{\mathrm{i}Ne^{\mathrm{i}Nt}}{e^{\mathrm{i}Nt} - \eta e^{2\pi\mathrm{i}\sigma}}$$

has residue one at each grid point. Integrate it around the rectangle with sides $\pm\pi \pm \mathrm{i}y + 2\pi\sigma/N$. It encloses exactly the N points in (6.1). On its vertical sides,

$$\mathcal{K}_\sigma(\pm\pi + \mathrm{i}v + 2\pi\sigma/N) = \frac{\mathrm{i}Ne^{-Nv}}{1 + e^{-Nv}}.$$

Thus no argument-dependent pole separation is assumed. The upper and lower horizontal expansions are respectively

$$-\mathrm{i}N \sum_{\ell \geq 1} \eta^\ell e^{-2\pi\mathrm{i}\ell\sigma} e^{\mathrm{i}\ell Nt}, \qquad \mathrm{i}N \left(1 + \sum_{\ell \geq 1} \eta^\ell e^{2\pi\mathrm{i}\ell\sigma} e^{-\mathrm{i}\ell Nt}\right).$$

Each term has an entire integrand, so its translated path can be moved to the fixed horizontal segment, retaining the short endpoint connectors. This is not a deformation of a meromorphic kernel through its moving poles.

The function $\sin t\, f(t) e^{\mathrm{i}nNt^2/(4\pi)}$ is odd. Reflect the upper side by $t \mapsto -t$, with its actual boundary orientation. Combining the two coefficients leaves

$$c_{\ell,N}(\sigma) = \eta^{\ell-1} U_{\ell-1}(\cos 2\pi\sigma), \qquad c_{1,N} = 1.$$

The zero-mode integral on the real segment is zero, but moving it there leaves its endpoint currents. The resulting nonzero-charge prefactor is $-N^2 C_{N,n}/(\pi\mathrm{i})$. In particular the reflected charges have already been paired; there will be no later unexamined sum of equal-action signs.

6.2 Infinite tails with real spectators

Let $A_0 = 1/400$, and use the heights

$$y_v = A_0 t_0 \frac{a_v}{|D_v|}. \tag{6.2}$$

In coordinate v, with all spectators real or on their physical translated grids, (4.2) bounds the exponent on the lower horizontal side by

$$F_0 - A_0 a_v t^2 + |b_v t y_v| - \ell y_v + 70 a_v y_v^2.$$

After maximizing t, the positive quadratic coefficient of y_v^2 is at most

$$70a_v + \frac{100 b_v^2}{a_v} \leq \frac{100|D_v|^2}{a_v}.$$

Since $L_v + 1 > t_0|D_v|$, every $\ell \geq L_v + 1$ loses at least

$$\tfrac{1}{2} A_0 t_0^2 a_v \geq \frac{1}{409600\pi}. \tag{6.3}$$

Further charges carry $e^{-N(\ell-L_v-1)y_v}$. The bound $|c_{\ell,N}(\sigma)| \le \ell e^{2\pi R'(\ell-1)}$ is absorbed for sufficiently large N, since this tree and $y_v > 0$ are fixed.

The original vertical sides have loss $\pi^2 a_v/400$. The chirp, finite-charge connector and imaginary Gaussian costs there are at most $\pi A_0 t_0 a_v$, $A_0 t_0^2 a_v$, and $70 A_0^2 t_0^2 a_v$. Their sum is less than half that endpoint loss. The zero-mode current obeys the same estimate. The translated-grid error $O(N^{-1})$ changes only the prefactor.

For completeness, define the finite real-integral functional

$$\mathcal{P}_v[f] = -\frac{N^2 C_{N,n_v}(\sigma_v)}{\pi \mathrm{i}} \sum_{\ell=1}^{L_v} c_{\ell,N}(\sigma_v) \int_{-\pi}^{\pi} \sin t \, f(t) e^{N(\mathrm{i}b_v t^2/2 - \mathrm{i}\ell t)} \, \mathrm{d}t.$$

Use the exact telescoping identity

$$\prod_v \mathcal{D}_v - \prod_v \mathcal{P}_v = \sum_j \left(\prod_{v<j} \mathcal{P}_v \right) (\mathcal{D}_j - \mathcal{P}_j) \left(\prod_{v>j} \mathcal{D}_v \right). \tag{6.4}$$

The left spectators are finite real integrals and the right spectators are physical grids. Each costs only a polynomial in N, with constants depending on fixed powers. Their grid heights are $O(N^{-1})$, not another coordinate's slope-dependent contour height. This is why the loss in (6.3) survives arbitrary relative degrees and twists.

Thus the exact retained contraction is, up to an error with that fixed loss, the finite sum

$$\begin{aligned} &\left(\prod_v -\frac{N^2 C_{N,n_v}(\sigma_v)}{\pi \mathrm{i}} \right) \sum_{1 \le \ell_v \le L_v} \left(\prod_v c_{\ell_v,N}(\sigma_v) \right) \\ &\quad \times \int_{[-\pi,\pi]^r} B_N(\boldsymbol{\tau}, s) \prod_v \sin \tau_v \, e^{N(\mathrm{i}\sum_v b_v \tau_v^2/2 - \mathrm{i}\sum_v \ell_v \tau_v)} \, \mathrm{d}\boldsymbol{\tau}. \end{aligned} \tag{6.5}$$

The *whole difference* is holomorphic in every formal σ_v on the unit disk. The original side has the holomorphic full crossing traces; the finite approximation has polynomial coefficients $c_{\ell,N}$. The maximum principle, one parameter at a time, extends the estimate from the circles to the disks. We can now set $\sigma_v = 0, s, -s$ as the actual types require. This step does not claim that each divided endpoint current separately has a removable singularity.

6.3 The connecting contours and the selected charge

Restrict the real integrals in (6.5) to $[-\rho,\rho]^r$, with $\rho = 1/8$. Their complement loses at least $1/(51200\pi)$ by (4.2); the real chirps and Fourier factors have modulus one. Translate this box to the imaginary part of its own critical point. On a connector write $\tau_v = t_v - \mathrm{i}v_v$, where $0 \le v_v \le -\operatorname{Im} \tau_{\ell,v}$. All these points lie in the radius-1/4 polydisk. Equation (4.4) and Young's inequality give

$$\operatorname{Re} \Phi_{\boldsymbol{\ell}}(\mathbf{t} - \mathrm{i}\mathbf{v}) \le F_0 - \tfrac{1}{8} \sum_v a_v t_v^2 + \sum_v \left[\tfrac{1+\varepsilon}{2} a_v + \frac{2b_v^2}{(3-4\varepsilon)a_v} \right] v_v^2 - \sum_v \ell_v v_v. \tag{6.6}$$

Indeed Young's inequality uses $(3-4\varepsilon)a_v t_v^2/8$ to absorb $b_v t_v v_v$. By (5.4),

$$\gamma_+ \max\{(1+\varepsilon)/2,\ 2/(3-4\varepsilon)\} < 3/4.$$

The last two sums in (6.6) are therefore at most $-\frac{1}{4}\sum_v \ell_v v_v$. Each connector has some $|t_v| = \rho$, and loses at least $1/(1024\pi)$. The negative Fourier terms of all spectators are retained in this calculation; replacing them by independent imaginary growth estimates would lose the result.

On the translated horizontal box the critical point has $|\operatorname{Re}\tau_v| < t_0/(1-\delta) < \rho$, and the real Hessian is strictly negative. Apply the holomorphic packet expansion (3.11) there. Taylor expansion at that critical point gives its Gaussian contribution; the rest has an exponential loss, and the odd Gaussian corrections vanish. For a nonminimal charge the original integral equals this saddle contribution plus an additive error at rate $F_0 - \eta$, where

$$\eta = \frac{1}{819200\pi}. \tag{6.7}$$

We do not assert a relative asymptotic for its original contour if its own action is smaller than that error. Equation (5.5) is what compares its saddle with the first one.

Every discarded source, edge-contour, real-box, tail and connector error has at least the loss (6.7), allowing polynomial prefactors. Finally

$$\beta \le \pi \frac{2550}{2401}\, 2^{-25} < \frac{1}{1638400\pi} = \frac{\eta}{2}. \tag{6.8}$$

All fixed-loss errors are strictly below the first-charge action. Each other retained positive charge is also strictly below it by (5.5). There are finitely many such charges for the fixed tree. These are two distinct comparisons, and both are needed.

7 The nonzero coefficient and the original knot volume

7.1 The coefficient of the whole invariant

Let

$$\mathcal{Q} = -\Psi''(\boldsymbol{\tau}_*) - \mathrm{i}\,\mathrm{diag}(b_v).$$

Its real quadratic part is positive on real vectors. Choose $\sqrt{\det \mathcal{Q}}$ by the complex Gaussian on that real tangent. This branch continues from positive definite real matrices in the domain with positive real part. Applying (3.11) to the first integral in (6.5) gives the explicit constant

$$D_{\mathcal{T},\mathbf{n}} = \left(-\frac{1}{\pi \mathrm{i}}\right)^r (2\pi)^{r/2} \frac{a(\boldsymbol{\tau}_*) \prod_v \sin \tau_{*,v}}{\sqrt{\det \mathcal{Q}}} \neq 0. \tag{7.1}$$

Each factor has already been proved nonzero. The charge-one coefficient is exactly one at every vertex, even at a colliding or constant grid. Hence no parity-dependent leading cancellation remains.

The powers of N can be read without an asymptotic normalization choice. The r exact trace factors supply N^{2r}. The e finite edge sums supply $N^{-3e/2}$, and the angular Gaussian supplies $N^{-r/2}$. Since $e = r - 1$, the total is $N^{3/2}$. At $s = 0$ all physical turn factors and H_N/W_k are one. Equations (2.4) and (6.1) therefore leave precisely

$$\vartheta_d^{-w(D)} \prod_v C_{N,n_v}(0) = q_N^{d^2(2w(D)+\sum_v n_v)/4} = \Omega_N.$$

Every final turn is included by the original compiler in (2.4); it is not removed in this count. This proves the complete leading term (1.4). All nonminimal charges and fixed-loss errors are absorbed using (5.5) and (6.8). The implied exponential rates and initial levels depend on the fixed tree and powers, as the theorem allows.

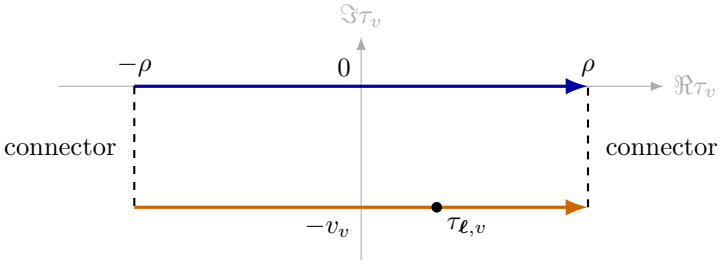


Figure 1: One coordinate of the finite angular deformation. The real box has endpoints $\pm\rho$, with $\rho = 1/8$; its translated horizontal side passes through the critical point. The two vertical sides are retained as connectors, with their boundary orientations when comparing the horizontal integrals. Here $v_v = -\Im\tau_{\boldsymbol{\ell},v} > 0$. Other coordinates can have intermediate downward displacements, so their negative Fourier terms are kept in (6.6). The earlier infinite-charge estimate uses real or translated-grid spectators, before this deformation.

7.2 Matching the geometric branch

The analytic action is exactly the action of [2, Theorem 1.1], not just a function with its complete Hessian. Its derivative (1.3) agrees on every edge with that theorem, and the constants agree: $2G_0 = v_{\mathrm{oct}}/\pi$, where v_{oct} is the regular ideal octahedron volume. This last equality also follows directly from the Lobachevsky integral at $\pi/4$.

Write $n_v = \epsilon_v - 2p_v$, with the retained parity $\epsilon_v \in \{-1, 0, 1\}$ in each actual disk frame. The geometric theorem fills the original primitive slope $\mu_v + p_v\lambda_v$, with periods

$$U_v = -2\pi\Psi_{\tau_v} - \mathrm{i}\epsilon_v\tau_v, \qquad V_v = 2\mathrm{i}\tau_v.$$

Its unit-charge equation is $\Psi_{\tau_v} + \mathrm{i}n_v\tau_v/(2\pi) = \mathrm{i}$, precisely (5.2) for $\boldsymbol{\ell} = \mathbf{1}$. The summed condition implies $\sqrt{d_v^2 + n_v^2} \geq 2^{25/2} > 630$. Furthermore (5.3) gives

$$|\tau_{*,v}| \leq \frac{2\pi}{(1-\delta)\sqrt{d_v^2 + n_v^2}} < 10^{-2}.$$

Thus our critical point lies in the geometric theorem's uniqueness chart. Its original filled metric is smooth, complete and finite-volume, including zero powers at high-degree vertices; every main cusp stays complete. Its absolute fundamental-chain calculation gives $\mathrm{Vol}(S^3 \setminus K_{\mathbf{n}}) = 2\pi\,\mathrm{Re}\,\mathcal{C}(\mathbf{1})$. There is no appeal here to a long-twist theorem for a zero power, or to a formal exponentiated slope without metric completion. Taking logarithms of the already nonzero (1.4) proves Theorem 1.1.

7.3 The centered main-meridian insertion

The finite scalar ledger of [1, Proposition 8.1] separates exactly $A_N(s)$ from the index-dependent factors H_N/W_k, the shifted crossing grids and the single closing factor $H_N(s)$. In the first-charge coefficient every grid parameter has disappeared because $c_{1,N}(\sigma) = 1$. The chirp has no remaining σ-dependence once its recorded scalar is extracted.

It remains to integrate the second line of (3.11) on the same joint contour. For any fixed holomorphic insertion $h(\boldsymbol{\tau})$, the normalized angular Gaussian evaluates it as $h(\boldsymbol{\tau}_*) + O(N^{-1})$: its linear Gaussian term vanishes, and the quadratic term and its interaction with the cubic phase contribute $O(N^{-1})$. The expansion to the next order and the fixed losses above make this statement uniform on $|s| \leq R$. Apply it to $-\pi s^2 \sum_e \cot(\pi x_e(\boldsymbol{\tau}))$, which already has the factor N^{-1}. The common first saddle correction cancels upon division by $F_N(1)$. The result

is (1.7) with a uniform $O_R(N^{-2})$ remainder. In particular the wire values are those at the filled critical point, not the complete values $1/4$.

Both A_N and H_N are nonzero and bounded away from zero on the fixed disk for large N. The ratio in (1.7) tends uniformly to one. This proves uniform eventual nonvanishing and the remaining assertions of Theorem 1.2. No division by an unproved nonzero finite invariant was used to establish the leading term.

8 Scope, dependencies and reproducibility

The condition is a theorem about every fixed literal marked tree satisfying (1.2). It is not uniform in a tree growing with N, and it is not a condition solely on the smallest crossing power. For example, in a star with d terminal vertices, central power zero and terminal powers n,

$$\mathfrak{S} = \frac{1}{d} + \frac{d}{1+n^2}.$$

Taking $d \geq 2^{27}$ and $|n| \geq \sqrt{2^{27}d}$ gives $\mathfrak{S} \leq 2^{-26}$. Knot applicability still requires the literal endpoint condition of Section 1; this arithmetic example does not supply that condition by itself. The proof uses neither the separate directed-message criterion nor the more general strict Conway grammar with bare junctions.

The mathematical dependencies have different jobs. The finite meridian calculus supplies the exact original invariant and all repeated-root data. The factorial lemma supplies one normalized holomorphic interpolant, and its application to this finite edge sum is proved in Section 3. Sections 4–6 prove the joint summed-condition analytic estimate, including the entire original source and Fourier period. Only after noncancellation is proved is the classical circle-tree theorem used to identify the action with metric volume. No quantum asymptotic is an input to that geometric theorem.

The residue and Gaussian steps belong to the classical contour method; [5, Sections 2.4(iii)–(iv)] gives background. Their contours and all losses relevant to this finite invariant have been specified above. From the repository root the paper-local reproduction command is

```
python3 papers/circle-tree-asymptotics/check.py \
    --output /tmp/circle-tree-check.json
```

It records source hashes and reuses the existing edge script's four exact symbolic identities and finite complex-product samples. It also checks the numerical budgets by exact rational interval arithmetic, using convergent series bounds for π, sinh and cosh. Negative controls reject altered budgets. Those finite calculations do not prove a uniform strip estimate, the charge tails or a volume theorem; the corresponding proofs are given here. The script and its recorded result are linked in the accompanying repository.

References

[1] *Exact Newton calculus for tangles with a moving meridian*, companion paper, Proposition 6.1, Theorem 7.1, and Proposition 8.1. Companion PDF.

[2] *Positive octahedra and primitive fillings for trees of crossing circles*, companion paper, Theorem 1.1. Companion PDF.

[3] *Holomorphic source asymptotics and volume limits for Conway closures*, companion paper, Lemma 2.1. Companion PDF.

[4] *Oscillatory filling sums with independently long twists*, companion paper. Companion PDF.

[5] NIST Digital Library of Mathematical Functions, Section 5.11, Gamma-function asymptotics, and Section 2.4, Contour integrals.